Choosing the Right Distribution for a Business Problem

Now that we know what **Normal, Binomial, and Poisson distributions** are, the next question is:

> **As a Data Scientist, how do you decide which distribution to use for a business problem?**

This is one of the first questions you'll ask before performing any statistical analysis or building a model.

Step 1: Understand the Business Problem

Before jumping into formulas, ask:

- What is the business trying to measure?
- Is it a measurement or a count?
- Are we counting events or successes?
- Is the data continuous or discrete?

Think like a business analyst first, not a statistician.

Step 2: Ask These Four Questions

```None
Business Problem
       │
       ▼
Is the variable Continuous?
       │
 ┌─────┴─────┐
 │           │
Yes         No
 │           │
 ▼           ▼
Normal    Is there a fixed
Distribution number of trials?
             │
       ┌─────┴─────┐
       │           │
      Yes         No
       │           │
       ▼           ▼
 Binomial     Are you counting
```

```
Distribution    events in time/
                space?
                     |
               ┌─────┴─────┐
               |           |
              Yes          No
               |
               ▼
     Poisson Distribution
```

Decision 1: Is the Data Continuous?

Continuous data can take **any value** within a range.

Examples:

- Height
- Weight
- Salary
- Delivery time
- Temperature

Business Example

Amazon wants to know:

> **What is the average delivery time?**

Possible values:

```None
25.4 minutes
27.8 minutes
30.2 minutes
29.6 minutes
```

Since time can take decimal values, it is **continuous**.

Choose: ✅ Normal Distribution

Decision 2: Is There a Fixed Number of Trials?

Suppose a marketing team sends promotional emails to **100 customers**.

Each customer either:

- Opens the email
- Doesn't open the email

The company asks:

> **How many customers will open the email?**

Notice:

- Fixed number of customers (100)
- Two outcomes
- Open / Not Open

This is a **Binomial Distribution** problem.

Decision 3: Are We Counting Events?

Uber wants to know:

> **How many ride requests arrive every minute?**

Possible answers:

```None
0
2
5
8
```

```
10
```

There is **no fixed number of trials**.

We are simply counting events occurring in a fixed time interval.

Choose: ✅ Poisson Distribution

Business Scenarios

Scenario 1 – Amazon

Business Question:

"What is the average delivery time?"

Variable

- Delivery Time

Type

- Continuous

Distribution :

Scenario 2 – Flipkart

Business Question

"What is the probability that exactly 18 out of 20 customers complete payment?"

Variable

- Completed payment

Possible Outcomes

- Success
- Failure

Distribution :

Scenario 3 – Swiggy

Business Question

"How many food orders arrive every minute?"

Variable

- Orders per minute

Distribution :

Scenario 4 – Hospital

Business Question

"What is the average waiting time?"

Distribution :

Scenario 5 – Manufacturing

Business Question

"Out of 500 products, how many are defective?"

Distribution :

Scenario 6 – Customer Support

Business Question

"How many complaint calls arrive every hour?"

Distribution :

Business Problem	Variable Type	Distribution
Delivery time	Continuous	Normal
Employee salary	Continuous	Normal
Product weight	Continuous	Normal
Height of students	Continuous	Normal

Number of customers who buy a product out of 100	Success/Failure	Binomial
Number of approved loans out of 500 applications	Success/Failure	Binomial
Number of defective products in a batch	Success/Failure	Binomial
Number of emails opened	Success/Failure	Binomial
Calls received per hour	Event Count	Poisson
Website visits per minute	Event Count	Poisson
Customer complaints per day	Event Count	Poisson
Accidents per month	Event Count	Poisson

Case Study Activity

Business Scenario	Distribution
Average monthly salary of employees	
Number of customers who purchased a premium subscription out of 200 visitors	
Number of support tickets received every hour	
Weight of apples in a supermarket	
Number of defective mobile phones in a batch of 500	
Number of ATM transactions per minute	

Daily sales revenue	
Number of successful loan approvals from 100 applications	
Number of flights arriving at an airport every hour	
Student exam marks	

Why Do We Need Probability Distributions?

Imagine you're hired as a Data Scientist at Amazon.

The manager asks:

> **"Can you predict tomorrow's sales?"**

You have millions of rows of data.

Can you simply look at the data and answer?

No.

You first need to understand **how the data behaves**.

That is exactly why we study **probability distributions**.

A distribution tells us **the pattern of the data**.

What Does a Distribution Tell Us?

A probability distribution helps answer questions like:

- Is the data continuous or discrete?
- Is the data symmetric or skewed?
- What values occur most often?
- How much variation is there?
- Are there any unusual (outlier) values?
- Can future values be predicted?
- Which statistical test or machine learning model should I use?

Without understanding the distribution, we are essentially making decisions without understanding the data.

Think of It Like a Doctor

Imagine a patient visits a doctor.

Does the doctor prescribe medicine immediately?

No.

The doctor first checks:

- Blood pressure
- Sugar level
- Temperature
- Heart rate

Only after understanding the patient's condition does the doctor prescribe treatment.

Similarly, in Data Science:

- **Data = Patient**
- **Distribution = Diagnosis**
- **Machine Learning Model = Treatment**

You should never choose a model before understanding the data.

Where Do We Use Distributions in Data Science?

1. Data Understanding (EDA)
2. Feature Engineering
3. Selecting the Right Machine Learning Model
4. Choosing the Right Statistical Test
5. Detecting Anomalies
6. Understanding Business Behavior

ML Perspective

Problem	Distribution	ML Task
House Price	Normal	Regression
Employee Salary	Normal	Regression
Delivery Time	Normal	Regression
Customer Churn (Yes/No)	Binomial	Classification
Fraud Detection	Binomial	Classification
Disease Prediction	Binomial	Classification
Customer Complaints	Poisson	Count Prediction

Website Visits	Poisson	Count Prediction
Orders Per Hour	Poisson	Forecasting/Poisson Regression

Key Takeaway

A common misconception is:

> **"Machine learning starts with choosing an algorithm."**

In reality, experienced data scientists start by asking:

- **What does my data look like?**
- **How is it distributed?**
- **What type of variable am I trying to model?**

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